

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Civil Engineering) SEMESTER - 6 Summer 2026 (Regular)

Course :Bachelor of Technology (Civil Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 6

Subject Code & Name: BTCVC603 - TRANSPORTATION ENGINEERING

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
4. Use of non-programmable scientific calculators is allowed.
5. Assume suitable data wherever necessary and mention it clearly.

Q1. Objective type questions. (Compulsory Question)

12

- 1 **Which road category primarily connects state capitals, major cities, and important economic centers across India?**
 - A. State Highway
 - B. National Highway
 - C. Major District Road
 - D. Village Road
- 2 **The main purpose of a reconnaissance survey in highway alignment is to:**
 - A. Prepare detailed drawings
 - B. Finalize pavement thickness
 - C. Identify feasible route alternatives
 - D. Conduct traffic volume studies
- 3 **In highway planning, O-D surveys are conducted primarily to determine:**
 - A. Soil characteristics
 - B. Travel patterns of road users
 - C. Pavement roughness
 - D. Roadside development
- 4 **Camber is primarily provided to:**
 - A. Improve aesthetics
 - B. Increase pavement thickness
 - C. Facilitate drainage of surface water
 - D. Reduce traffic congestion
- 5 **The aggregate property evaluated by the Los Angeles Abrasion Test is:**
 - A. Shape
 - B. Toughness
 - C. Hardness and abrasion resistance
 - D. Water absorption

- 6 **The Plasticity Index of soil is:**
 A. $LL + PL$
 B. $LL - PL$
 C. $PL - LL$
 D. $SL - PL$
- 7 **Ductility test on bitumen indicates its:**
 A. Softening temperature
 B. Viscosity
 C. Adhesion property
 D. Ability to deform before breaking
- 8 **The 85th percentile speed is generally considered:**
 A. Minimum speed on the road
 B. Average speed of all vehicles
 C. Speed exceeded by 85% of vehicles
 D. Speed below which 85% of vehicles travel
- 9 **Which road sign category includes "No Entry" signs?**
 A. Informatory Signs
 B. Warning Signs
 C. Mandatory Signs
 D. Direction Signs
- 10 **The coefficient of friction used in stopping sight distance calculations generally:**
 A. Increases with vehicle speed
 B. Is independent of pavement condition
 C. Decreases with increase in speed
 D. Depends only on traffic volume
- 11 **A rotary intersection becomes unsuitable when:**
 A. Traffic flow is balanced
 B. Traffic volume is very high and weaving becomes excessive
 C. Approach roads are four in number
 D. Design speed is low
- 12 **The modulus of subgrade reaction (k-value) is an important parameter in:**
 A. Flexible pavement design
 B. Marshall Mix Design
 C. Rigid pavement design
 D. Traffic volume studies

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Q2. Solve the following.

- A) What are the factors that control the alignment of road? 6
- B) State and explain different types of road pattern with diagram. 6

Q3. Solve the following.

- A) The speed of overtaking and overtaken vehicles are 60 & 40 kmph respectively on a two way traffic road. If the acceleration of overtaking vehicle is 0.80 m/s^2 . (i) Calculate safe overtaking sight distance. (ii) Mention the minimum length overtaking zone (Assume $t = 2$ Seconds)
- B) Define: a) Right of way b) Road Margin c) Carriage way d) Shoulder e) Kerb f) Camber 6

- Q4. Solve Any Two of the following.
- A) Explain CBR test also state its advantages & limitation. 6
 - B) i. Define Flakiness Index and Elongation Index ii. Explain the procedure for Aggregate Impact Test. 6
 - C) In the Marshall method of mix design the coarse aggregate, fine aggregate, filler and bitumen having respective specific gravities of 2.62, 2.72, 2.70 & 1.02 are mixed in the ratio 55, 34, 6, 4.8 and 5.6 percent respectively. What will be the theoretical specific gravity of the mix? 6
- Q5. Solve Any Two of the following.
- A) Explain the methods used for the analysis and interpretation of traffic survey data. 6
 - B) What is Traffic Engineering? Explain traffic characteristics. 6
 - C) Write a short note on Road Markings and road Signs. 6
- Q6. Solve Any Two of the following.
- A) Explain different components of pavement with its function? 6
 - B) Give the comparison between rigid and flexible pavement in detail? 6
 - C) Provide an overview of other modes of transportation such as railways, airways, waterways, and pipeline transportation. 6

*** End ***